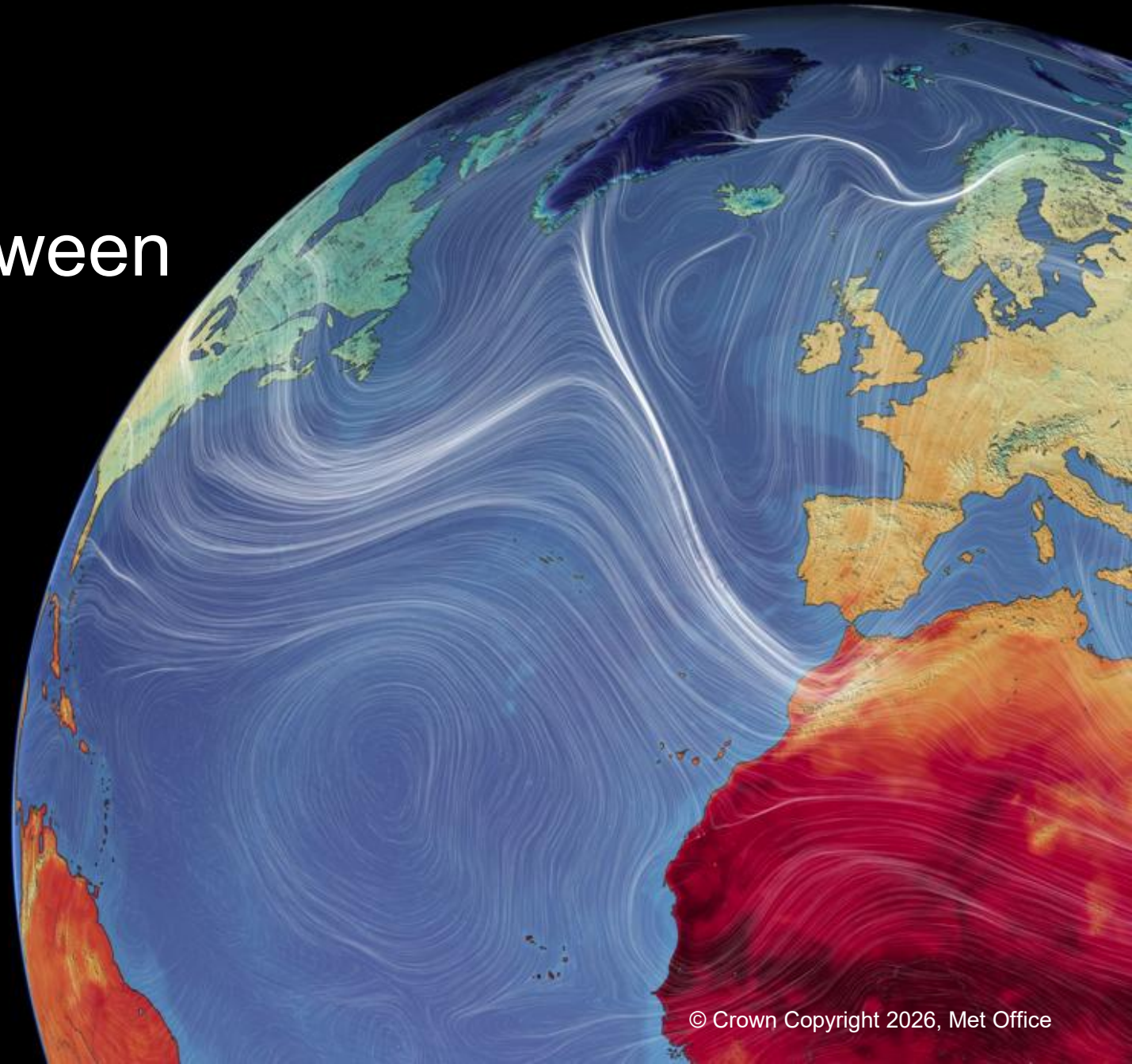


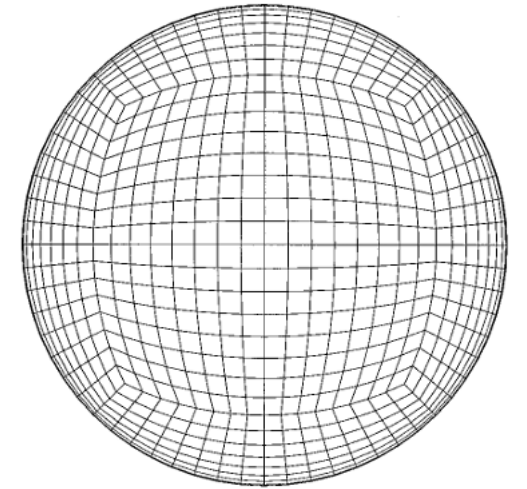
Interpolation, NWP, and Some Points In Between

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Foundation Scientific Software
Engineer (Met Office)



Why is Interpolation?

- Initial state and output of numerical simulation (NWP) are “on-grid.”
- **Grid to Point:**
 - Observed data from recording stations not at grid-points.
 - Used for o-b statistic in Data Assimilation.
- **Grid to Grid:**
 - Changing resolution or grid-type (lon-lat, cubed-sphere,...).
 - Used for transforming to spectral representation



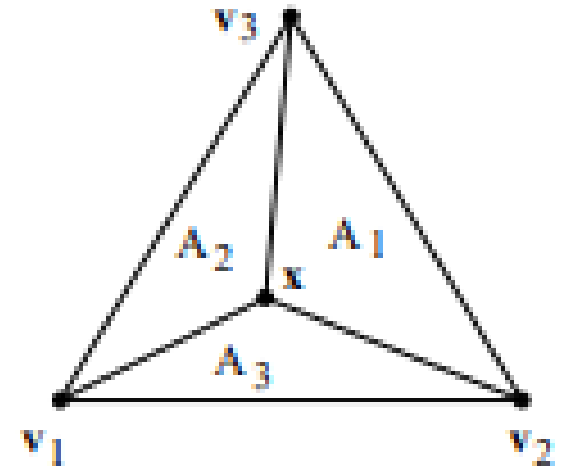
Cubed-Sphere Grid
Ronchi, Iacono, Paolucci
J. Comp. Phys 124, 93-114
(1996)

What is Interpolation?

- Consider a point x in triangle $\Delta(v_1, v_2, v_3) \in \mathbb{R}^3$
- From v_1 , we can reach x by going along $v_2 - v_1$ then $v_3 - v_1$, so

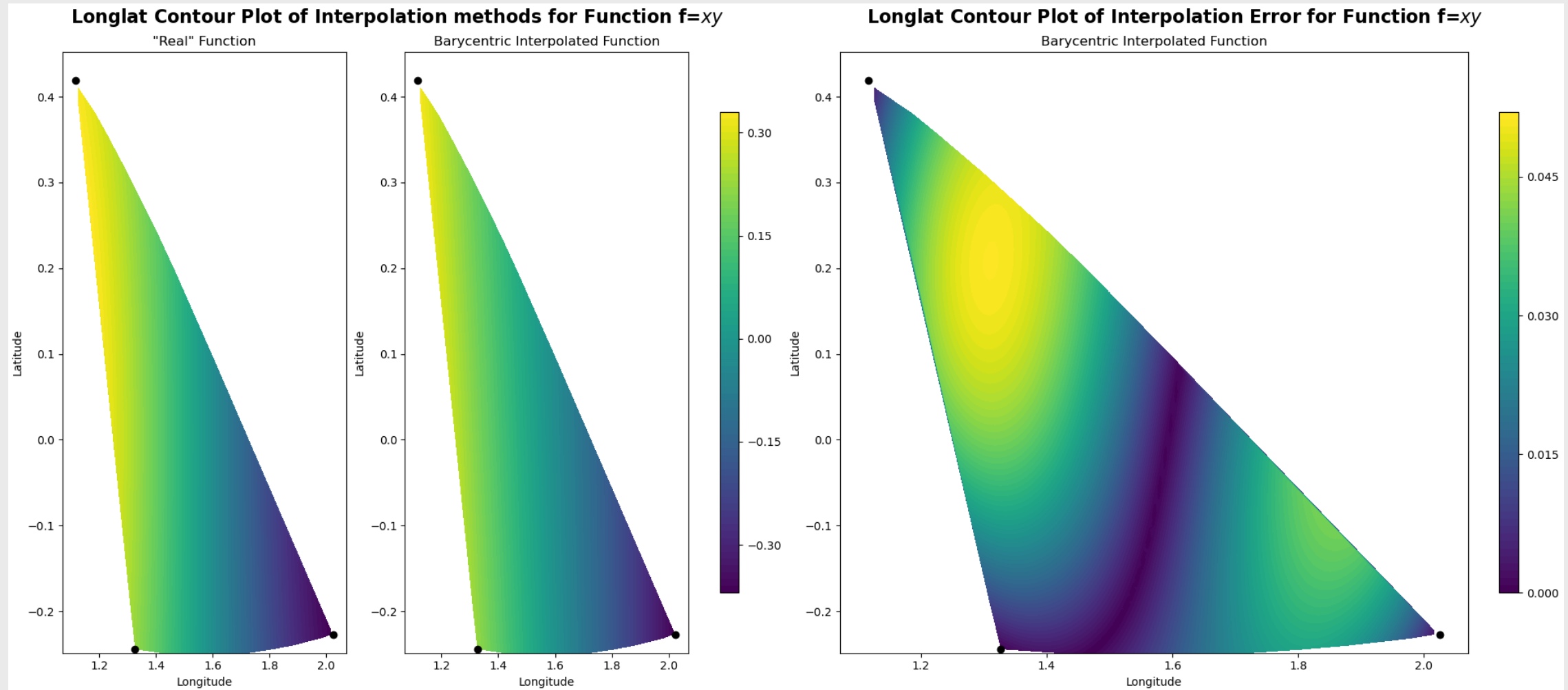
$$\begin{aligned}x &= v_1 + \mu(v_2 - v_1) + \nu(v_3 - v_1) \\&= (1 - \mu - \nu)v_1 + \mu v_2 + \nu v_3 \\&:= \phi_1 v_1 + \phi_2 v_2 + \phi_3 v_3\end{aligned}$$

- Any point in triangle can be **uniquely** written as convex combo of vertices, where coefficients sum to 1.
- Given $f: \mathbb{R}^3 \rightarrow \mathbb{R}$, with known values at vertices, there exists **interpolant** $g(x) := \sum_{i=1}^3 \phi_i(x) f(v_i)$ such that $g(v_i) = f(v_i)$



Coefficients can be related to areas of sub-triangles.
Floater. Acta Numerica (2016)

What is Interpolation?



What isn't Interpolation?

- Attempt to do the same for a quad in $\mathbb{R}^3$:

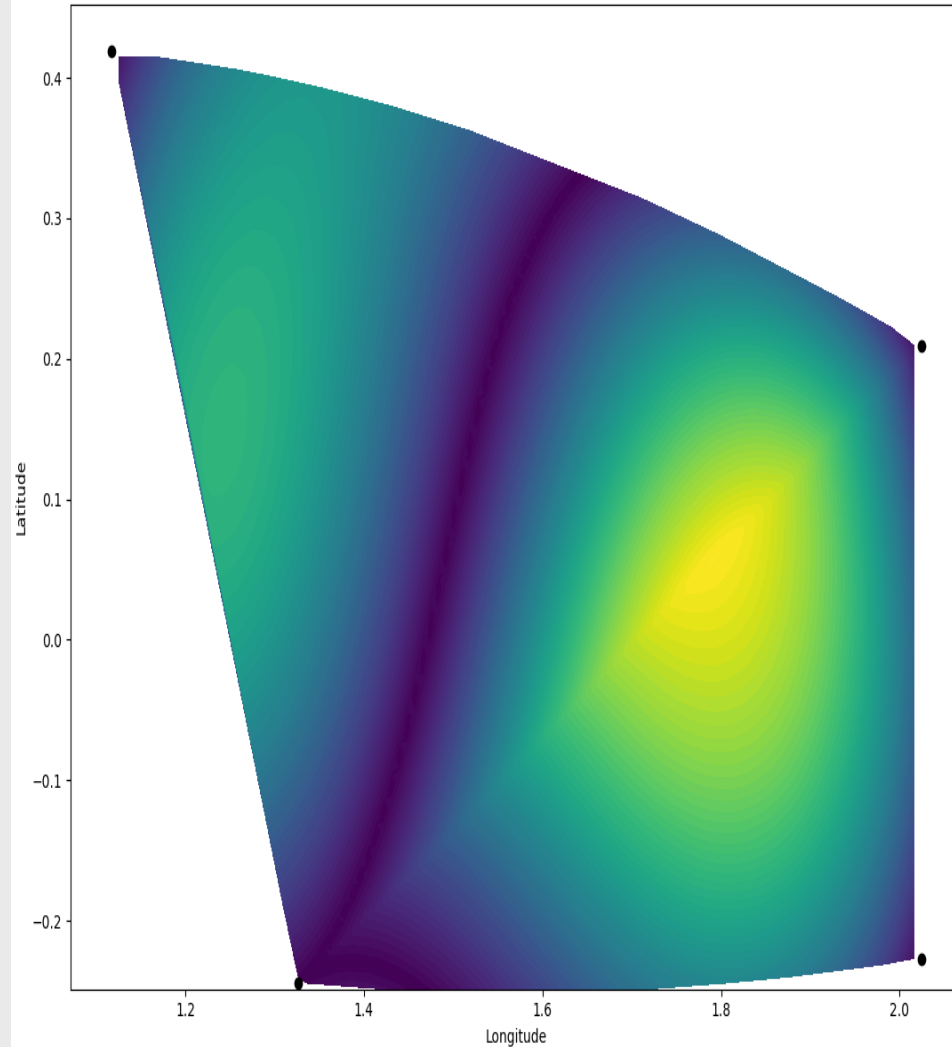
$$\mathbf{x} = \phi_1 \mathbf{v}_1 + \phi_2 \mathbf{v}_2 + \phi_3 \mathbf{v}_3 + \phi_4 \mathbf{v}_4$$

- Have 4 unknowns and 2 or 3 independent equations.
Even imposing coefficients sum to 1, still underdetermined – **not unique!**
- We have a choice of interpolant:
 - Many software packages use bilinear interpolation.
 - Involves dividing quad into triangles (arbitrary, not invariant, can be discontinuous).
 - Can find something more suitable using non-uniqueness!

What isn't Interpolation?

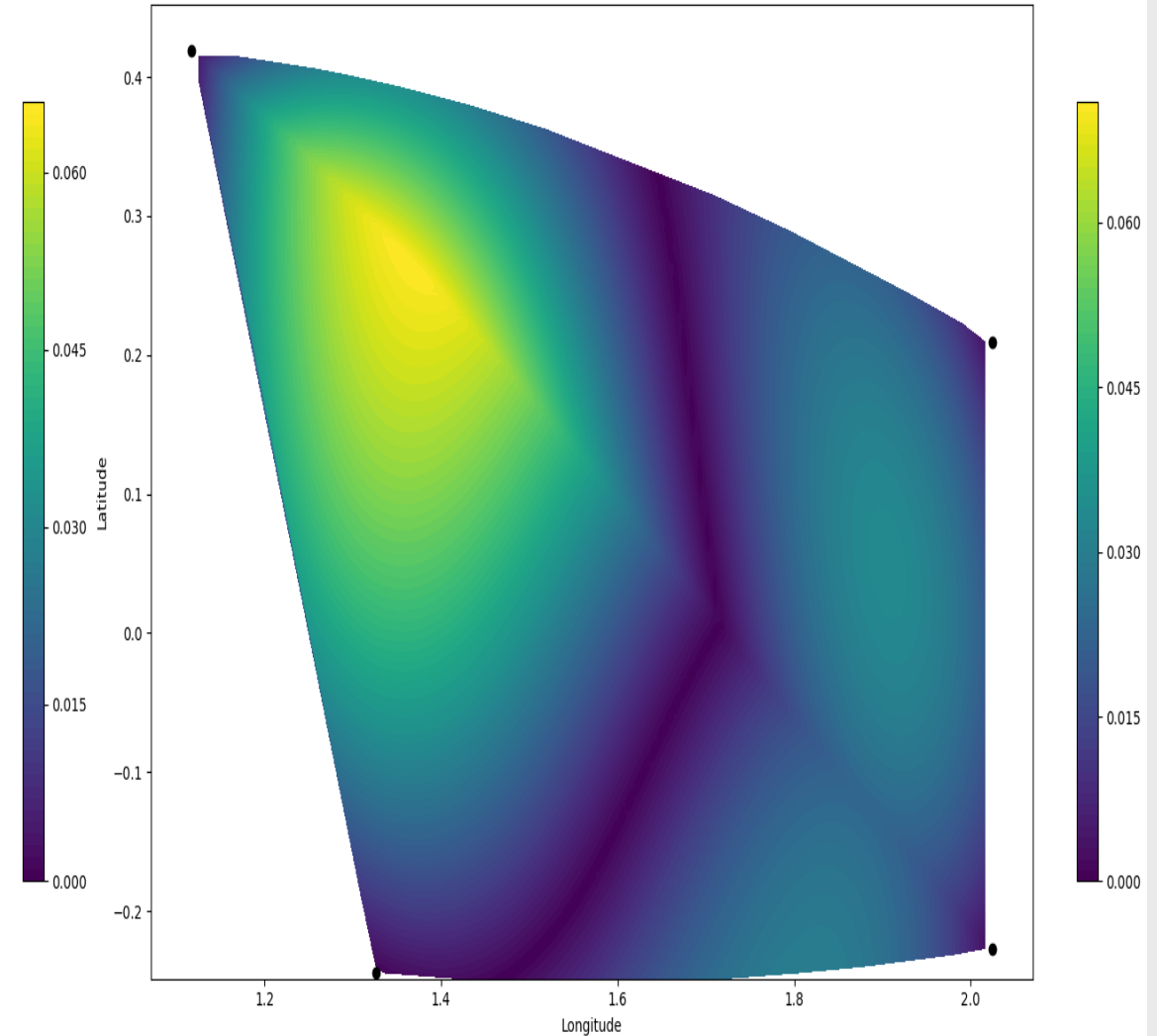
Longlat Contour Plot of Interpolation Error for Function $f=xy$

Barycentric Interpolated Function



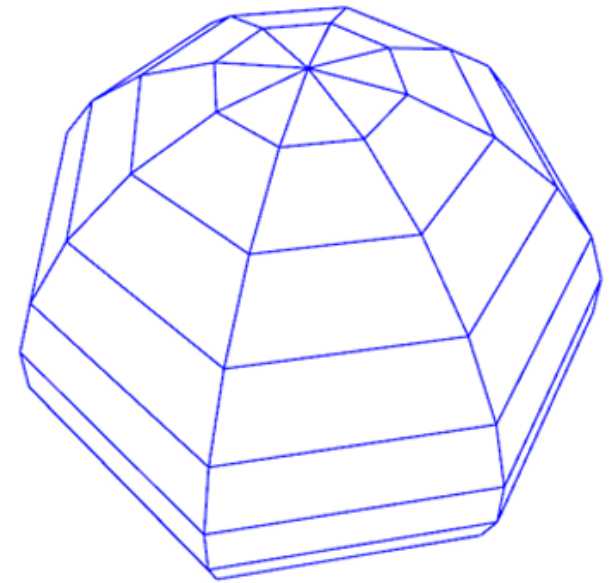
Longlat Contour Plot of Interpolation Error for Function $f=xy$

Barycentric Interpolated Function



What isn't Interpolation? (2)

- Recall $x = (1 - \mu - \nu)v_1 + \mu v_2 + \nu v_3$
- This is linear: e.g. along side from v_1 to v_2 , we get equation of straight line
$$x = (1 - \mu)v_1 + \mu v_2$$
- Spherical polygons get “flattened” and gridded sphere gets “faceted” like a die, so not accurately represented.
- Good spherical method would describe circular arc between pairs of points instead.



Example of a faceted sphere.
*Bloomenthal, J. Comp.
Graphics Tech. 11, 3 (2022)*

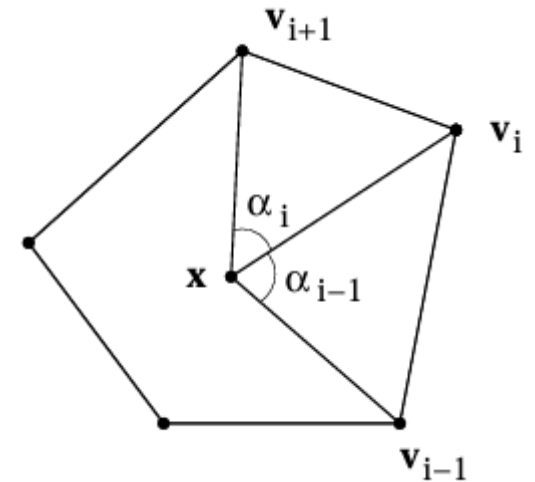
How is Interpolation?

- Solution: accurate & invariant interp. on arbitrary **spherical** polygons.
- **Spherical Mean Value** interpolation
[Floater, Acta Numerica (2016)] (now in ECMWF/Atlas)

$$\phi_i = \frac{w_i}{\sum_{j=1}^n w_j v_j \cdot x}, \quad w_i = \frac{\tan\left(\frac{\alpha_i}{2}\right) + \tan\left(\frac{\alpha_{i-1}}{2}\right)}{\|v_i \times x\|},$$

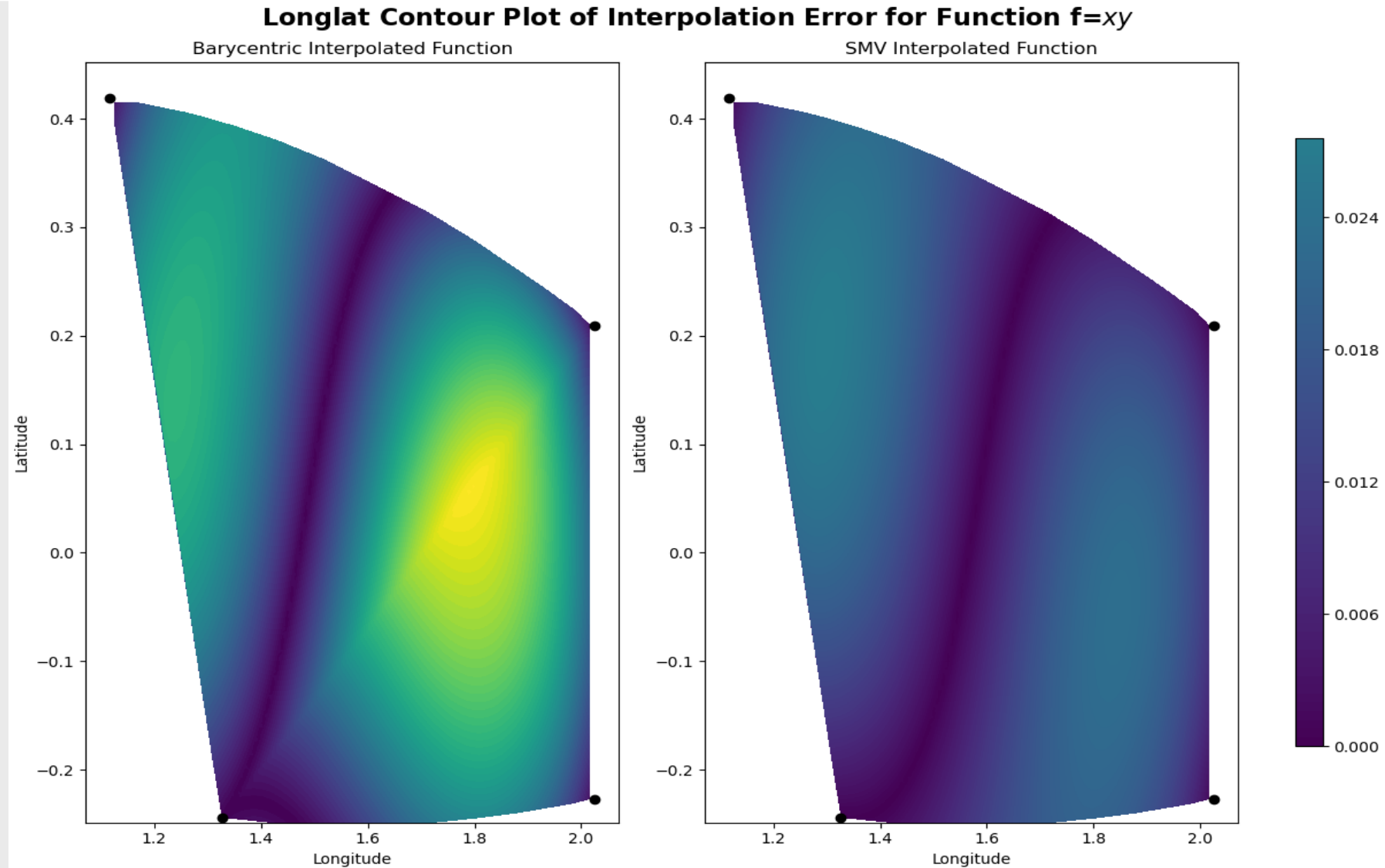
$$g(x) := \sum_{i=1}^n \phi_i(x) f(v_i)$$

- Trig bad computationally – tractable when written as dot/cross products of x and v_i .
- Note: These sum to ≥ 1 !



Coefficients can be related to
angles about x .
Floater. Acta Numerica
(2016)

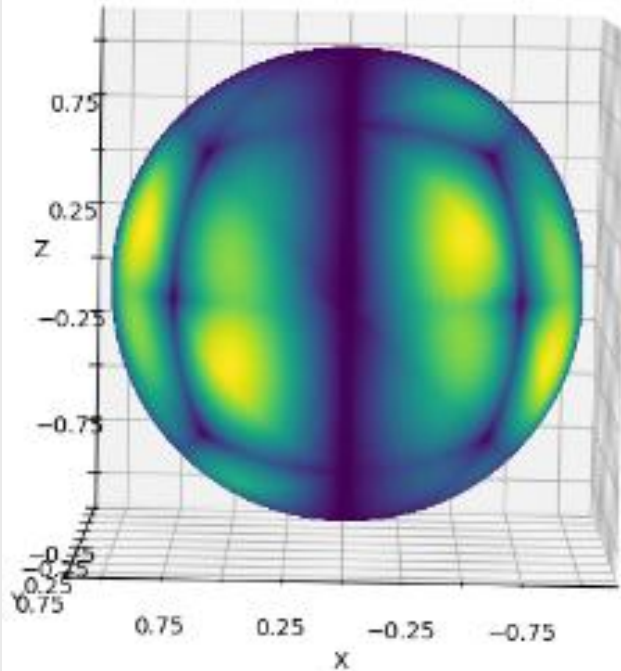
How is Interpolation?



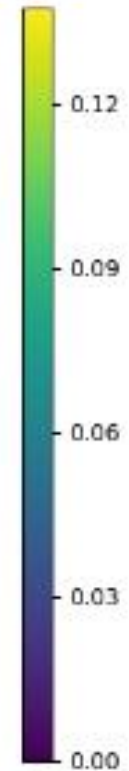
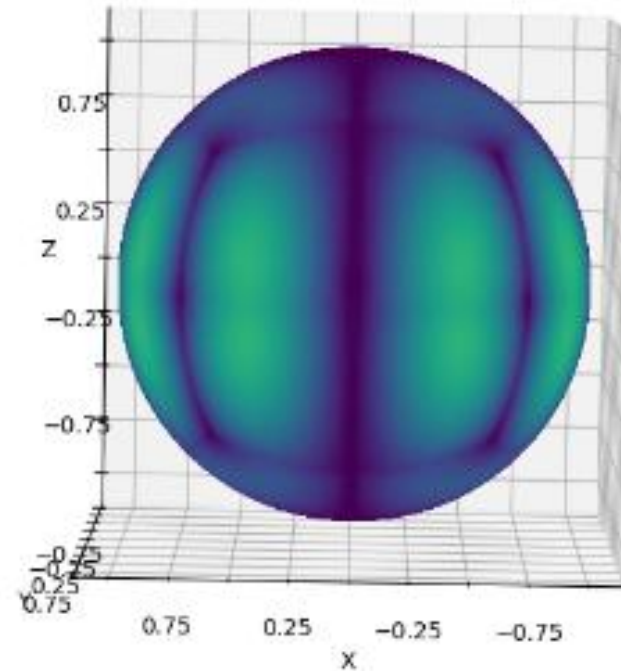
How is Interpolation?

Comparison of Interpolation Error on Spherical Polygons for Function $f=xy$

Barycentric Interpolated Function



SMV Interpolated Function



Who is Interpolation?

- M. Floater “Generalized barycentric coordinates and applications” *Acta Numerica* (2016)
- W. E. Deconinck “Atlas: A library for numerical weather prediction and climate modelling” *Comp. Phys. Comm.* vol. 220 (2017)
- C. Ronchi, R. Iacono, P.S. Paolucci “The Cubed Sphere: A New Method for the Solution of PDEs in Spherical Geometry” *J. Comp. Phys.* Vol 224 (1996)
- J. Bloomenthal “Barycentric Quad Rasterization” *J. Comp Graphics Tech.* Vol. 11 (2022)